# 1.3 Experiments and Observational Studies

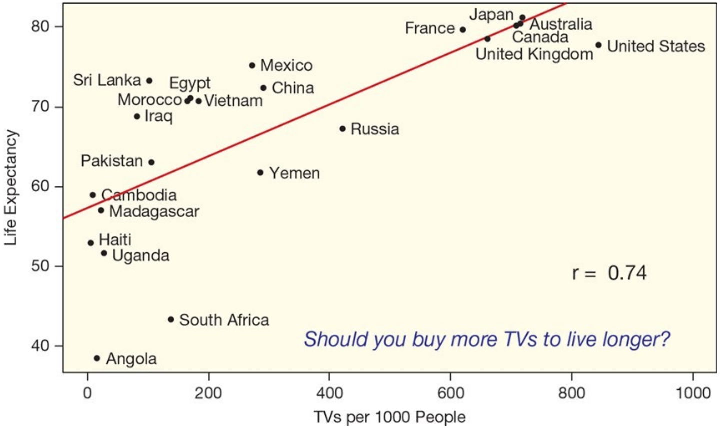
- Outline
  1. Association versus Causation
  2. Confounding variables
  3. Observational studies vs Experiments
  4. Randomized Experiments

# Association and Causation

- **Association**: Two variables are associated if values of one variable tend to be related to values of the other variable.
- **Causation**: Two variables are causally associated if changing the value of the explanatory variable influences the value of the response variable.

# Example 1.6 | TVs and Life Expectancy

- Should we buy another TV to live longer?




Source: Lock[5] (2021), John Wiley & Sons, Inc.

- Note: **Association is not necessarily causal**.

# Exercise 1.4 | Explanatory, Response, and Causation

- For each of the following three headlines,
  A. Identify the **explanatory** and **response** variables (if appropriate).
  B. Whether or not the headline implies a **causal** association.

1. "*Daily Exercise Improves Mental Performance*"

2. "*Want to lose weight? Eat more fiber!*"

3. "*Cat owners tend to be more educated than dog owners*"

- Other examples of association but not causation (i.e., spurious correlation) at `https://www.tylervigen.com/spurious-correlations`.

# Exercise 1.5 | Explanatory, Response, and Causation

- Note: **Association is not necessarily causal**.
- Come up with two variables that are **associated**, but **not causally**.
- Come up with two variables that are **causally** associated.

  1. Which is the explanatory variable?
  2. Which is the response variable?

# Confounding Variables

- A third variable that is associated with both the explanatory variable and the response variable is called a **confounding variable**, also known as a **confounding factor** or **lurking variable**.
  1. A confounding variable can offer a plausible explanation for an association between the explanatory and response variables.
  2. **Whenever confounding variables are present (or may be present), a causal association cannot be determined.**
  3. Conceptualization of explanatory, response, and confounding variables:

# Example 1.6 (revisited) | TVs and Life Expectancy

- Should we buy another TV to live longer?
- If not, identify a **confounding variable** or provide a plausible explanation for the relation.

# Exercise 1.6 | Confounding Variable

- For each of the following relationships, identify a possible **confounding variable**:
  1. More ice cream sales have been linked to more deaths by drowning.
  2. The total amount of beef consumed and the total amount of pork consumed worldwide are closely related over the past 100 years.
  3. People who own a yacht are more likely to buy a sports car.
  4. Air pollution is higher in places with a higher proportion of paved ground relative to grassy ground.
  5. People with shorter hair tend to be taller.

# Observational Study vs Experiment

- An **observational study** is a study in which the researcher does not actively control the value of any variable, but simply observes the values as they naturally exist.
  - Example: Based on observations you make in nature, you suspect that female crickets choose their mates on the basis of their health. Then, observe health of male crickets that mated.
- An **experiment** is a study in which the researcher actively controls one or more of the explanatory variables.
  - Example: Deliberately infect some males with intestinal parasites and see whether females tend to choose healthy rather than ill males.

# Exercise 1.7 | Observational Study vs Experiment

- In order to investigate whether women are more likely than men to prefer Democratic candidate, a political scientist selects a large sample of registered voters, both men and women. She asks every voter whether they voted for the Republican or the Democratic candidate in the last election. This is
  1. an experiment
  2. an observational study

- A market research company wishes to find out whether the population of students at NAU prefers brand A or brand B of instant coffee. A random sample of students is selected, and each one is asked to try brand A and then brand B, or vice versa (with the order determined at random). They then indicate which brand they prefer. This is an example of
  1. an experiment
  2. an observational study

# Exercise 1.7 | Observational Study vs Experiment

- To determine if smoking causes cancer, researchers surveyed a large sample of adults. For each adult they recorded whether the person had smoked regularly at any period in their life and whether the person had cancer. They compared the proportion of cancer cases in those who had smoked regularly at some time in their lives with the proportion of cases in those who had never smoked regularly at any point in their lives. The researchers found there was a higher proportion of cancer cases among those who had smoked regularly than among those who had never smoked regularly. This study is
  1. an experiment
  2. an observational study

- Some people believe that exercise raises the body's metabolic rate for as long as 12 to 24 hours, enabling us to continue to burn off fat after our workout has ended. In a study of this effect, subjects walked briskly on a treadmill for several hours. Their metabolic rates were measured before, immediately after, and 12 hours after the exercise. This study is
  1. an experiment
  2. an observational study

# Randomized Experiments

- There are almost always confounding variables in observational studies.
- **In general, observational studies are not used to establish causation.**
- How can we effectively avoid confounding effects?
- The answer is **randomization**: Allocate the experimental units to treatments **at random (by chance)**.
- In a **randomized experiment**, the explanatory variable for each unit is determined randomly, before the response variable is measured.

# Terminology for Experiments

- The individuals in an experiment are the **experimental units**. If they are human, we call them **subjects**.
- **Factors** are the explanatory variables in an experiment.
  - If the experiment involves giving two different doses of a drug, we are testing two **levels** of the factor (drug effect).
- **Treatment** is a specific experimental condition applied to experimental units.
  - Each treatment is formed by combining a specific level of each of the factors.

# Terminology for Experiments

- **Nuisance variables (lurking Variables)** other variables which influence the response variable but which are not of interest.
- A response to a treatment is **statistically significant** if it is larger than you would expect by chance (due to random variation among the subjects). We will learn how to determine this later.
- Two variables are **confounded** when their effects on a response variable cannot be distinguished from each other.
  1. Example: If we simply observe cell phone use and brain cancer, any effect of radiation on the occurrence of brain cancer is confounded with lurking variables such as age, occupation, and place of residence.
  2. Well designed experiments take steps to defeat confounding.

# Exercise 1.8 | Terminology for Experiments

- A group of college students believes that herbal tea has remarkable restorative powers. To test its theory, the group makes weekly visits to a local nursing home, visiting with residents, talking with them, and serving them herbal tea. After several months, many of the residents are more cheerful and healthy. The explanatory variable in this experiment is
  1. the emotional state of the residents
  2. herbal tea
  3. the fact that this is a local nursing home
  4. the college students
- Referring to the previous question, the lurking variable in the experiment is
  1. the emotional state of the residents
  2. herbal tea
  3. the fact that this is a local nursing home
  4. visits of the college students

# Exercise 1.8 | Terminology for Experiments

- Sickle-cell disease is a painful disorder of the red blood cells that affects mostly blacks in the United States. To investigate whether the drug *Hydroxyurea* can reduce the pain associated with sickle-cell diseases, a study by the National Institute of Health gave the drug to 150 sickle-cell sufferers and a placebo to another 150. The researchers then counted the number of episodes of pain reported by each subject.
  1. Experimental units (or subjects):
  2. Treatment(s):
  3. Factor(s):
  4. Response variable(s):

# Exercise 1.8 | Terminology for Experiments

- Ability to grow in shade may help pines found in the dry forests of Arizona resist drought. How well do these pines grow in shade? Investigators planted pine seedlings in a greenhouse in either full light or light reduced to 5% of normal by shade cloth. At the end of the study, they dried the young trees and weighted them.

  1. Experimental units (or subjects):
  2. Treatment(s):
  3. Factor(s):
  4. Response variable(s):

# Exercise 1.8 | Terminology for Experiments

- New varieties of corn with altered amino acid patterns may have higher nutritive value than standard corn, which is low in the amino acid lysine. An experiment compares two new varieties, called opaque-2 and floury-2, with normal corn. Corn-soybean meal diets using each type of corn are prepared at three different protein levels, 12%, 16%, and 20%, giving nine diets in all. Researchers assign 10 one-day-old male chicks to each diet and record their weight gains after 21 days. The weight gain of the chicks is a measure of the nutritive value of their diet.

  1. Experimental units (or subjects):
  2. Treatment(s):
  3. Factor(s):
  4. Response variable(s):

# Three Principles of Experiments

- **Control** the effects of lurking variables on the response, simply by comparing two or more treatments.
- **Randomize** subjects. Allocate the experimental units to treatments at random (by chance).
- **Replicate** each treatment on enough subjects to reduce chance variation in the results.

# Principle 1 | Control (Comparison)

- Experiments are comparative in nature. We compare the response to a treatment to:
  1. another treatment
  2. no treatment (a control)
  3. a placebo
  4. or any combination of the above
- A **control** is a situation where no treatment is administered. It serves as a reference mark for an actual treatment (e.g., a group of subject does not receive any drug or pill of any kind).
- A **placebo** is a fake treatment, that resembles the active treatment as much as possible.
  1. Using a placebo is only helpful if participants do not know whether they are getting the placebo or the active treatment.
  2. If possible, randomized experiments should be **double-blinded**: neither the subjects nor the experimenter know which individuals got which treatment until the experiment is completed. The goal is to avoid forms of placebo effects and biases based on interpretation.

# Example 1.7 | Gastric freezing

- **Initial design**. “Gastric freezing” is a clever treatment for ulcers in the upper intestine. The patient swallows a deflated balloon with tubes attached, then a refrigerated liquid is pumped through the balloon for an hour. The idea is that cooling the stomach will reduce its production of acid and so relieve ulcers. An experiment showed that gastric freezing did reduce acid production and relieve ulcer pain. The treatment was safe and easy and was widely used for several years.
- What do you think? Is this experiment reasonable?

# Example 1.7 | Gastric freezing

- This gastric freezing experiment has several **critiques**:
  1. This experiment was poorly designed because it tells *little* thing about whether the response was due to the treatment (gastric freezing) or to *lurking* variables.
  2. The patients' response may have been due to the *placebo effect*. Many patients respond favorably to any treatment, even a placebo. This may be due to trust in the doctor and expectations of a cure, or simply to the fact that medical conditions often improve without treatment.
  3. The experiment conducted above gave misleading results because the effects of gastric freezing were *confounded* (or *mixed up*) with the placebo effect.

# Example 1.7 | Gastric freezing

- **Improved design**. A later experiment divided ulcer patients into two groups. One group was treated by gastric freezing as before. The other group received a placebo treatment in which the liquid in the balloon was at body temperature rather than freezing.
  - The results: 34% of the 82 patients in the treatment group improved, but so did 38% of the 78 patients in the placebo group.
  - This showed that gastric freezing was *no better* than a placebo.
- Notes:
  1. **Control group**: the group of experimental units that received a placebo treatment. This *controls* the effects of *outside* variables on the outcome.
  2. Now the reason why we could say gastric freezing was no better than a placebo is because the *only* difference between the two groups is the actual effect of gastric freezing. Other possible *lurking* variables were evenly considered on both groups).

# Principle 2 | Randomization

- The experiment is **biased** if it systemically favors certain outcomes.
- The best way to exclude biases from an experiment is to **randomize** the design.
- **Randomization** allocates the experimental units to treatments *at random* (*by chance*).
- Randomization makes the treatment groups probabilistically *alike* on all *lurking variables* (or *nuisance factors*), thereby avoiding systematic *bias*.

# Example 1.8 | Fertilizers and crop yield

- Two fertilizers are compared for their effects on crop yield. Suppose a field made up of 24 plots is divided so that the 12 easternmost plots receive fertilizer A and the remaining plots receive fertilizer B.
  - If there is a trend in soil quality, the soil quality will be confounded with the treatment.
  - If we randomly assign the treatments to plots the distribution of soil quality should be approximately the same in the A plots and as in the B plots.
- Notes:
  1. Randomization will tend to neutralize all lurking variables.
  2. Differences in average crop yield will be due to the fertilizers.

# Randomization Methods

- How to Randomize?
  1. Use a table of random digits: less popular lately.
  2. Generate random numbers using a statistical software or a random number generating app.
- Example: In the fertilizer example, we should divide 24 plots into two groups of 12 plots. Use a random number generating software to perform this randomization.
  1. Step 1. Label: Give each plot a numerical label, starting from 1 to 24.
  2. Step 2. Random number generator: Generate twelve random numbers in the range of 1 to 24 without replacement using a random number (to avoid same numbers to be selected repeatedly) generating software.
  3. The plots associated with the selected random numbers form the fertilizer A group. The non-selected plots are the fertilizer B group.

# Principle 3 | Replication

- The best way to make sure your conclusions are robust is to **replicate** your treatments on another subjects – do it over.
- **Replication**
  1. ensures that particular results are *not due to uncontrolled factors or errors of manipulation*.
  2. *reduces chance variation* in the results.

# Exercise 1.9 | Effect of a new medication on allergy

- Researchers wish to determine if a new experimental medication will reduce the symptoms of allergy sufferers without the side effect of drowsiness. To investigate this question, the researchers give the new medication to 50 adult volunteers who suffer from allergies. Forty four of these volunteers report a significant reduction in their allergy symptoms without any drowsiness. This study could be improved by
  1. including people who do not suffer from allergies in the study in order to represent a more diverse population.
  2. repeating the study with only the 44 volunteers who reported a significant reduction in their allergy symptoms without any drowsiness, and giving them a higher dosage this time.
  3. using a control group.
  4. all of the above.

# Exercise 1.10 | Green tea and prostate cancer

- A study was conducted on 60 men with prostatic intraepithelial neoplasia (PIN) lesions, some of which turn into prostate cancer. Half of these men were randomized to take 600 mg of green tea extract daily, while the other half were given a placebo pill The study was double-blind, neither the participants nor the doctors knew who was actually receiving green tea. After one year, only 1 person taking green tea had gotten cancer, while 9 taking the placebo had gotten cancer.
  1. Are the three principles of experiments considered in this experiment?
  2. A difference this large is unlikely to happen just by random chance. Can we conclude that green tea really does help prevent prostate cancer?

# Two Types of Randomized Experiments

- An experimental is called **completely randomized** or a **randomized comparative experiment** if all experimental units are allocated *at random among all treatments*.
- In a **matched pairs experiment**, we choose
  1. pairs of subjects that are closely matched (e.g., same gender, height, weight, age, and race). Within each pair, we randomly assign who will receive which treatment.
  2. a single subject, and give the two treatments to this subject over time in random order. In this case, the "matched pair" is just the same subject at different points in time.
- Notes:
  1. Matched pairs experiments always compare *two* treatments.
  2. The order of the treatments in a matched pairs experiment can influence the unit's response, so we *randomize the order* for each unit (eg. by a coin toss).
  3. Matched pairs experiments can produce more *precise* results than randomized comparative experiments.

# Randomized Comparative Experiment

- Example. A food company assesses the nutritional quality of a new “instant breakfast” product by feeding it to newly weaned male white rats and measuring their weight gain over a 28-day period. A control group of rats receives a standard diet for comparison. The researchers use 30 rats for the experiment and so must divide them into two groups of 15. To do this in a completely randomized fashion, they put the cage numbers of the 30 rats in a hat, mix them up, and draw 15. These rats form the experimental group and the remaining 15 make up the control group. The following figure outlines the design of this experiment.

# Randomized Comparative Experiment

- A graphical illustration of the experiment:


1. The use of randomization makes the two groups similar in all respects before the treatments are applied.
2. Differences in average weight gain of the two groups must be due to the diets.
3. If there is only small number of rats in each group, the difference in average weight gain of the two groups is only due to the diets?

# Matched Pairs Experiment

- Example. A foreign language institute hosted 20 French teachers for 4 weeks. At the beginning of the period, the teachers were given the Modern Language Association's listening test of understanding of spoken French. After 4 weeks of immersion in French in and out of class, the listening test was given again. (The actual French spoken in the two tests were different, so that simply taking the first test should not improve the score on the second test.)

| Teacher | Pretest | Posttest | Teacher | Pretest | Posttest |
|---|---|---|---|---|---|
| 1 | 32 | 34 | 11 | 30 | 36 |
| 2 | 31 | 31 | 12 | 20 | 26 |
| 3 | 29 | 35 | 13 | 24 | 27 |
| 4 | 10 | 16 | 14 | 24 | 24 |
| 5 | 30 | 33 | 15 | 31 | 32 |
| 6 | 33 | 36 | 16 | 30 | 31 |
| 7 | 22 | 24 | 17 | 15 | 15 |
| 8 | 25 | 28 | 18 | 32 | 34 |
| 9 | 32 | 26 | 19 | 23 | 26 |
| 10 | 20 | 26 | 20 | 23 | 26 |

# Example 1.9 | Randomized Comparative vs. Matched Pairs

- Are cereal leaf beetles more strongly attracted by the color yellow or by the color green? Agriculture researchers want to know, because they detect the presence of the pests in farm fields by mounting sticky boards to trap insects that land on them. The board color should attract beetles as strongly as possible. We must design an experiment to compare yellow and green by mounting boards on poles in a large field of oats.
  1. Design a randomized comparative experiment of this study.
  2. Design a matched pairs experiment of this study.

## Example 1.9 | Randomized Comparative vs. Matched Pairs

- A randomized comparative experiment of this study can be conducted as follows:
  1. The experimental units are locations within the field far enough apart to represent observations. We erect a pole at each location to hold the boards.
  2. We randomly select half the poles to receive a yellow board while the remaining poles receive green.
  3. The locations vary widely in the number of beetles present. For example, the alfalfa that borders the oats on one side is a natural host of the beetles, so locations near the alfalfa will have extra beetles.
  4. This variation among experimental units can hide the systemic effect of the board color.

# Example 1.9 | Randomized Comparative vs. Matched Pairs

- A matched pairs experiment of this study can be conducted as follows:
  1. We mount boards of both colors on each pole. Then two boards on the same pole form a **pair**.
  2. Because the boards are mounted one above the other, we select the color of the top board at random. Just toss a coin for each board – for example, if the coin falls heads, the yellow board is mounted above the green board.
  3. The observations (numbers of beetles trapped) are matched in pairs from the same poles. We compare the number of trapped beetles on a yellow board with the number trapped by the green board on the same pole.
  4. It is more efficient to employ a matched pairs design in this experiment.

# Exercise 1.11

- Researchers wish to determine if a new experimental medication will reduce the symptoms of allergy sufferers without the side effect of drowsiness. To investigate this question, the researchers give the new medication to 50 adult volunteers who suffer from allergies. Forty four of these volunteers report a significant reduction in their allergy symptoms without any drowsiness. This study could be improved by
  1. including people who do not suffer from allergies in the study in order to represent a more diverse population.
  2. repeating the study with only the 44 volunteers who reported a significant reduction in their allergy symptoms without any drowsiness, and giving them a higher dosage this time.
  3. using a control group.
  4. all of the above.

# Exercise 1.11 (continued)

- 100 volunteers who suffer from severe depression are available. 50 are selected at random and are given a new drug which is thought to be effective in treating severe depression. The other 50 are given an existing drug for treating severe depression. A psychiatrist evaluates the symptoms of all volunteers after four weeks in order to determine if there has been substantial improvement in the severity of the depression. The factor in this study is
    1. which treatment the volunteers receive.
    2. the extent to which the depression was reduced.
    3. the use of a psychiatrist to evaluate the severity of depression.
    4. the use of randomization and the fact that this was a comparative study.
- This study will be double blind if
    1. neither drug had any identifying marks on it.
    2. neither the volunteers nor the psychiatrist knew which treatment any person had received.
    3. all volunteers were not allowed to see the psychiatrist nor the psychiatrist allowed to see the volunteers during the session during which the psychiatrist evaluated the severity of the depression.
    4. all of the above.

# Exercise 1.11 (continued)

- Will a fluoride mouthwash used after brushing reduce cavities? 20 sets of twins were used to investigate this question. One member of each set of twins used the mouthwash after each brushing, the other did not. After six months, the difference in the number of cavities of those using the mouthwash was compared with the number of those who did not use the mouthwash. This experiment
  1. uses placeboes.
  2. uses double blinding.
  3. is a randomized comparative experiment.
  4. is a matched pairs experiment.

# Exercise 1.11 (continued)

- Is the right hand generally stronger than the left in right-handed people? You can crudely measure hand strength by placing a bathroom scale on a shelf with the end protruding, then squeezing the scale between the thumb below and the four fingers above it. The reading of the scale shows the force exerted. Design a matched pairs experiment to compare the strength of the right and left hands, using 15 right-handed people as subjects. Use a coin to do the required randomization.